

WORTHINGTON, MN, USA

WMO: 726587

Lat: 43.645N

Lon: 95.580W

Elev: 1570

StdP: 13.88

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94927

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-7.6	-20.4	1.9	-5.5	-16.1	2.4	-3.3	36.0	14.8	33.0	17.3	11.3	280	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	88.7	72.8	85.9	71.4	82.5	69.5	76.5	84.6	74.6	82.4	72.6	80.1	14.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	132.0	81.5	72.2	126.2	80.5	70.2	117.6	78.5	41.2	84.4	39.2	82.4	37.3	79.8	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.9	27.0	24.5	DB	-17.0	94.1	4.6	3.1	-20.3	96.3	-23.0	98.1	-25.7	99.8	-29.0	102.1	(4)
(5)			WB	-17.8	79.5	4.5	2.6	-21.0	81.4	-23.7	82.9	-26.2	84.4	-29.5	86.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.3	15.8	20.0	32.1	45.4	56.9	67.6	71.3	68.4	61.7	48.5	33.8	20.9		
	DBStd	21.87	13.05	12.73	12.51	10.00	9.02	7.05	5.82	5.62	8.50	9.83	11.23	11.80		
	HDD50	4252	1059	840	565	198	35	0	0	0	10	147	495	903		
	HDD65	7757	1524	1260	1019	591	284	52	14	30	162	518	935	1368		
	CDD50	2550	0	0	12	60	248	529	661	570	360	100	10	0		
	CDD65	578	0	0	0	3	32	131	210	135	61	6	0	0		
	CDH74	5008	0	0	2	55	375	1195	1809	975	517	79	1	0		
	CDH80	1403	0	0	0	12	121	382	527	220	124	17	0	0		
(14) Wind	WSAvg	12.0	13.4	13.7	13.5	13.9	13.3	11.2	9.1	8.4	10.0	12.0	13.1	12.9		
(15) Precipitation	PrecAvg	28.6	0.6	0.7	1.3	2.8	3.6	4.7	3.4	3.3	3.1	2.0	1.1	0.7		
	PrecMax	43.6	1.5	1.5	3.4	7.3	6.5	11.2	9.3	9.6	9.2	5.2	3.6	2.3		
	PrecMin	18.3	0.0	0.1	0.0	0.4	1.0	1.9	0.4	0.6	0.4	0.2	0.0	0.1		
	PrecStd	5.9	0.4	0.4	0.9	1.6	1.4	1.9	2.2	2.3	2.6	1.5	1.1	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	54.9	70.3	81.2	90.2	92.2	91.4	89.7	88.2	81.9	68.3	52.2		
		MCWB	38.3	47.6	55.3	61.0	66.9	72.0	76.8	76.5	70.4	63.0	53.8	46.5		
	2%	DB	38.8	46.1	62.8	72.6	82.3	88.1	88.4	84.5	83.7	74.7	61.0	45.1		
		MCWB	34.4	41.4	52.5	55.7	63.6	70.1	75.5	73.0	69.6	58.9	50.7	39.4		
	5%	DB	36.1	40.3	54.9	66.2	78.6	84.1	85.2	82.0	80.7	69.7	54.9	38.9		
		MCWB	33.1	36.4	47.2	52.3	61.6	68.1	73.7	71.0	66.3	57.3	46.5	34.1		
	10%	DB	33.6	36.4	48.4	61.5	72.5	81.2	82.2	79.5	76.4	63.9	49.6	36.1		
		MCWB	31.1	33.4	43.1	49.7	58.6	66.1	71.4	69.0	64.9	53.8	43.1	32.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	49.2	58.8	62.6	70.4	76.4	80.4	78.1	74.2	67.2	57.1	47.4		
		MCDB	45.2	54.6	64.9	77.8	83.9	86.4	87.1	85.8	82.6	76.2	65.8	49.9		
	2%	WB	35.2	41.5	53.0	58.2	67.0	73.4	77.5	75.3	71.4	62.4	51.6	39.1		
		MCDB	38.4	46.2	61.1	69.6	78.4	83.2	85.6	82.9	79.7	69.9	58.3	44.5		
	5%	WB	33.0	37.0	48.5	54.6	63.6	70.7	75.5	73.4	69.1	58.7	47.5	35.0		
		MCDB	35.6	40.4	55.1	63.6	74.5	81.0	83.3	80.1	77.5	67.1	54.5	38.5		
	10%	WB	30.8	33.3	43.3	51.3	60.4	68.4	73.7	71.1	66.8	54.8	43.2	32.5		
		MCDB	32.7	36.0	49.7	59.5	69.1	78.2	80.9	77.7	74.1	63.4	49.5	35.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	15.9	16.2	19.7	19.8	19.0	18.8	18.5	20.9	19.8	16.8	14.6		
		MCDBR	17.5	18.4	24.9	28.7	27.5	23.3	20.2	20.3	24.3	27.1	24.4	18.3		
	5% WB	MCWBR	14.3	14.5	16.6	16.3	13.4	11.0	11.2	12.1	12.9	15.1	16.1	14.2		
		MCWBR	16.6	18.0	24.2	25.4	23.0	20.4	18.5	20.3	23.2	23.0	23.0	17.1		
(40) Clear-Sky Solar Irradiance	taub		0.262	0.278	0.309	0.362	0.383	0.398	0.408	0.407	0.361	0.322	0.286	0.267		
	taud		2.446	2.423	2.437	2.320	2.333	2.348	2.308	2.285	2.389	2.486	2.512	2.459		
	Ebn at Noon		281	294	297	286	282	277	272	269	274	273	269	267		
	Edh at Noon		23	28	31	38	39	39	40	40	33	26	22	21		
(44) All-Sky Solar Radiation	RadAvg		598	944	1224	1514	1752	1953	2093	1731	1383	929	630	468		
	RadStd		60	89	106	137	126	132	80	128	95	118	44	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page